

Bài tập lần 7

Tài liệu xony-T.L.pdf, trang 18, 19

Bài 3.

$$a) \begin{cases} u_{tt} = 9u_{xx} \quad (*) \\ u(x,0) = \chi_{[-1,1]}(x) \\ u_t(x,0) = x \cdot \chi_{[1,2]}(x) = \begin{cases} x & \text{ khi } 1 \leq x \leq 2 \\ 0 & \text{ khi ngược lại} \end{cases} \end{cases}$$

Phương trình đặc trưng.

$$(dx)^2 - 9(dt)^2 = 0$$

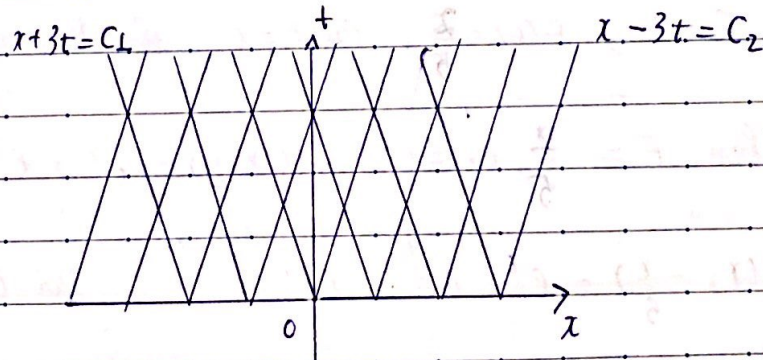
$$\Leftrightarrow dx \pm 3dt = 0$$

$\Rightarrow (*)$ là Hyperbolic

$$dx \pm 3dt = 0$$

$$\Rightarrow \begin{cases} x + 3t = C_1 \\ x - 3t = C_2 \end{cases}$$

$$x - 3t = C_2$$



Đổi biến $\begin{cases} \xi = x+3t \\ \eta = x-3t \end{cases}$

		v_ξ	v_η	$v_{\xi\xi}$	$v_{\xi\eta}$	$v_{\eta\eta}$
0	u_x	-1	1	0	0	0
0	u_t	3	-3	0	0	0
-9	u_{xx}	0	0	1	2	1
0	u_{xt}	0	0	3	0	3
1	u_{tt}	0	0	9	-18	9
Tổng	0	0	0	0	-36	0

→ Dạng chính tắc của (*) là

$$-36 v_{\xi\eta} = 0$$

$$\Leftrightarrow v_{\xi\eta} = 0$$

$$\rightarrow v_{\xi} = g(\xi)$$

$$\rightarrow v = F(\xi\lambda + G(\eta)) = F(\eta) + G(\xi)$$

$$\rightarrow u(x,t) = F(x-3t) + G(x+3t)$$

$$u(x,0) = F(x) + G(x) = \chi_{[-1;1]}(x)$$

$$u_t(x,t) = -3F'(x-3t) + 3G'(x+3t)$$

$$\rightarrow u_t(x,0) = -3F'(x) + 3G'(x) = x \chi_{[1;2]}(x)$$

Ta có hệ
$$\begin{cases} u(x,0) = F(x) + G(x) = \chi_{[-1;1]}(x) = f(x) & (1) \\ u_t(x,0) = -3F'(x) + 3G'(x) = x \chi_{[1;2]}(x) = g(x) & (2) \end{cases}$$

Đặc biệt $u) \rightarrow F'(x) + G'(x) = f'(x)$

Thay vào (2) Ta có

$$-3(f'(x) - G'(x)) + 3G'(x) = g(x)$$

$$\rightarrow G'(x) = \frac{1}{6} (3f'(x) + g(x))$$

$$\therefore G(x) = \frac{1}{2} f(x) + \frac{1}{6} \int_0^x g(s) ds + C_1$$

$$F(x) = f(x) - G(x) = \frac{1}{2} f(x) - \frac{1}{6} \int_0^x g(s) ds - C_1$$

$$\rightarrow u(x,t) = \frac{1}{2} f(x-3t) - \frac{1}{6} \int_0^{x-3t} g(s) ds - C_1 + \frac{1}{2} f(x+3t) + \frac{1}{6} \int_0^{x+3t} g(s) ds + C_1$$

$$= \frac{f(x-3t) + f(x+3t)}{2} + \frac{1}{6} \int_{x-3t}^{x+3t} g(s) ds$$

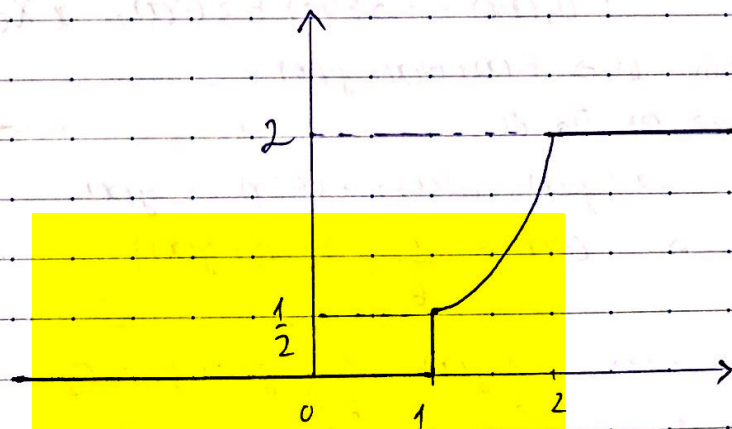
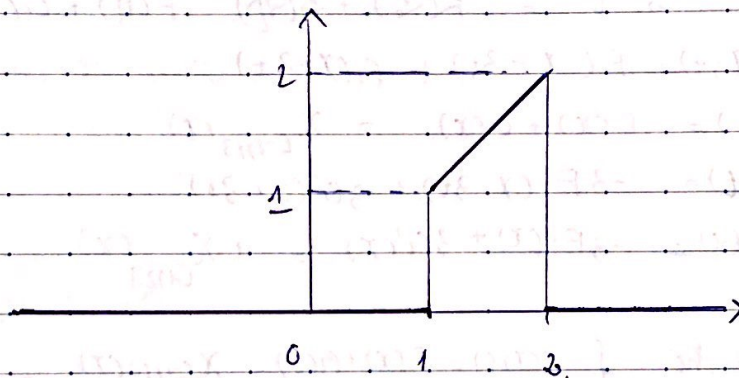
Thử lại
$$\begin{cases} u(x,0) = f(x) = \chi_{[-1;1]}(x) \\ u_t(x,0) = g(x) = x \chi_{[1;2]}(x) \end{cases} \quad (T/M)$$

$$u_{tt} = 9u_{xx}$$

Sóng đều: $F(x) = \frac{f(x)}{2} - \frac{1}{6} \int_0^x g(s) ds$

Sóng đều: $G(x) = \frac{f(x)}{2} + \frac{1}{6} \int_0^x g(s) ds$

$$g(x) = \chi_{[1,2]}(x) = \begin{cases} x & \text{nếu } 1 \leq x \leq 2 \\ 0 & \text{nếu ngược lại} \end{cases}$$



$$\int_0^x g(s) ds = \begin{cases} 0 & \text{nếu } x \leq 1 \\ \frac{x^2}{2} & \text{nếu } 1 < x \leq 2 \\ 2 & \text{nếu } x > 2 \end{cases}$$

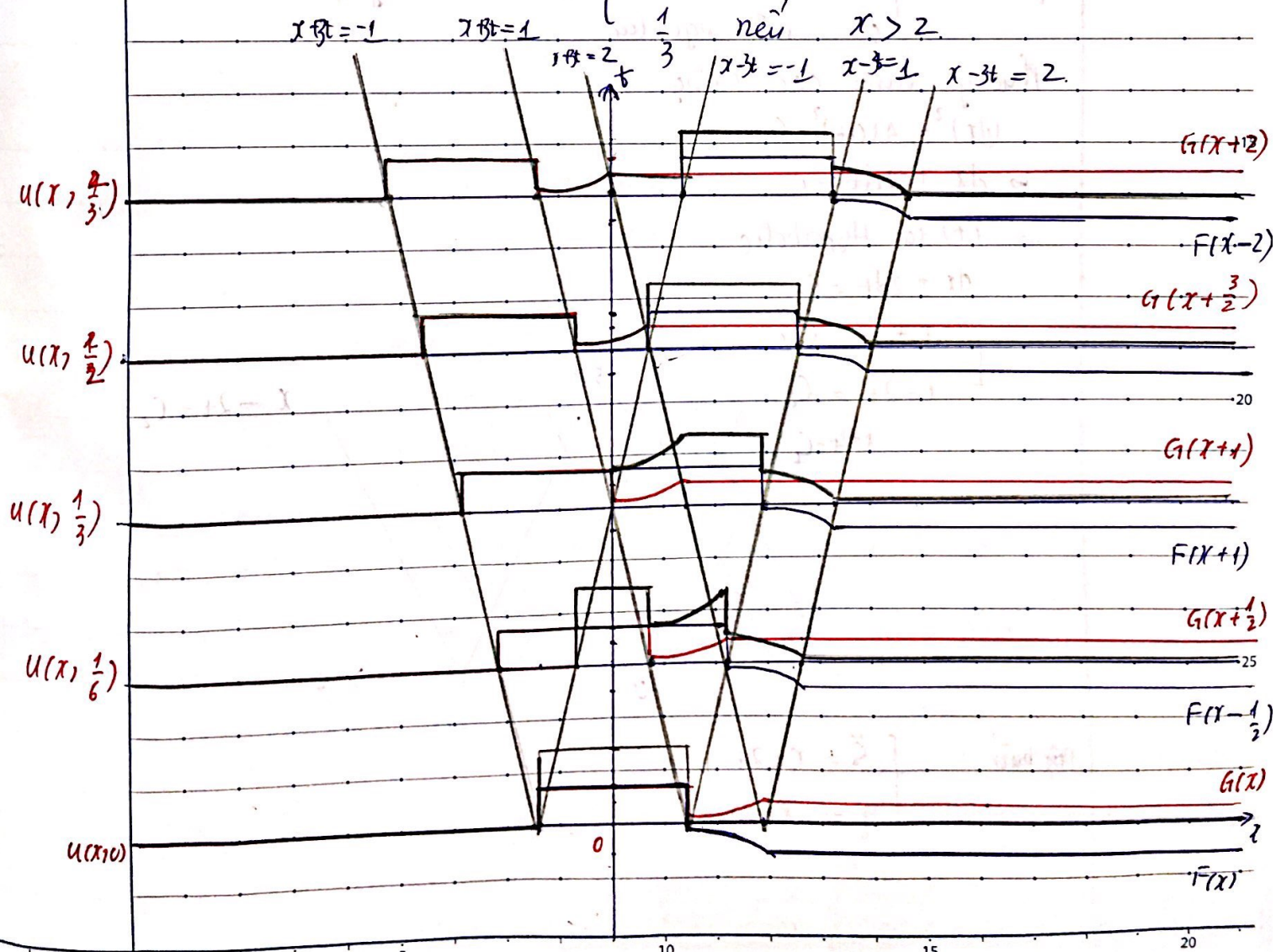
$$f(x) = \chi_{[-1,1]}(x) = \begin{cases} 1 & \text{nếu } -1 \leq x \leq 1 \\ 0 & \text{nếu ngược lại} \end{cases}$$

$$\rightarrow F(t) = \frac{f(x)}{2} - \frac{1}{6} \int_0^x g(x) ds =$$

$$\begin{cases} 0 & \text{nếu } x < -1 \\ \frac{1}{2} & \text{nếu } -1 < x < 1 \\ -\frac{x^2}{12} & \text{nếu } 1 < x < 2 \\ -\frac{1}{3} & \text{nếu } x > 2 \end{cases}$$

$$G(x) = \frac{f(x)}{2} + \frac{1}{6} \int_0^x g(x) ds =$$

$$\begin{cases} 0 & \text{nếu } x < -1 \\ \frac{1}{2} & \text{nếu } -1 < x < 1 \\ \frac{x^2}{12} & \text{nếu } 1 < x < 2 \\ \frac{1}{3} & \text{nếu } x > 2 \end{cases}$$



Tập các đ' lẻ ch' của $u(x,t)$ gồm 8 tr'.

$$\left\{ \begin{array}{l} x+3t = +1 \\ x+3t = 2 \\ x-3t = +1 \\ x-3t = 2 \end{array} \right.$$

Sách Poincaré - Rubinstein Trang 93-97

4.3

$$u_t - 4u_{xx} = 0, \quad x \in \mathbb{R}, t > 0. \quad (*)$$

$$u(x,0) = \begin{cases} 1-x^2 & \text{nếu } -1 < x < 1 \\ 0 & \text{nếu ngược lại} \end{cases} = f(x)$$

$$u_t(x,0) = \begin{cases} 4 & \text{nếu } 1 < x < 2 \\ 0 & \text{nếu ngược lại} \end{cases} = g(x)$$

Phương trình đặc trưng

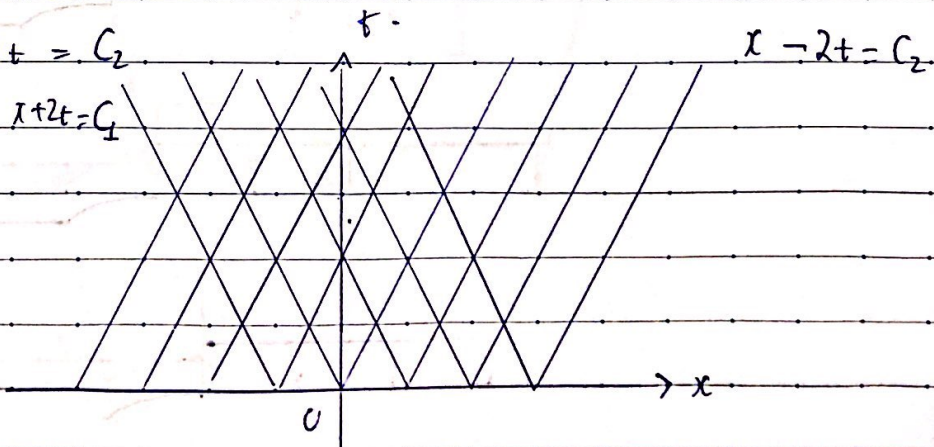
$$(dx)^2 - 4(dt)^2 = 0$$

$$\Rightarrow dx \pm 2dt = 0$$

$\Rightarrow (*)$ là Hyperbolic

$$dx \pm 2dt = 0$$

$$\Rightarrow \begin{cases} x+2t = C_1 \\ x-2t = C_2 \end{cases}$$



Đổi biến

$$\begin{cases} \xi = x+2t \\ \eta = x-2t \end{cases}$$

		v_{ξ}	v_{η}	$v_{\xi\xi}$	$v_{\xi\eta}$	$v_{\eta\eta}$
0	u_x	1	1	0	0	0
0	u_t	2	-2	0	0	0
-4	u_{xx}	0	0	1	2	1
0	u_{xt}	0	0	2	0	-2
1	u_{tt}	0	0	4	-8	4
Tổng	0	0	0	0	-12	0

→ Dạng chính tắc của (1) là

$$-12 v_{\xi\eta} = 0$$

$$\Leftrightarrow v_{\xi\eta} = 0$$

$$\Rightarrow v_{\xi} = g(\xi)$$

$$\Rightarrow v = F(\eta) + G(\xi)$$

$$\Rightarrow u(x,t) = F(x-2t) + G(x+t)$$

$$u(x,0) = F(x) + G(x) = f(x)$$

$$u_t(x,0) = -2F'(x) + 2G'(x) = g(x)$$

$$\Rightarrow u_t(x,0) = -2F' + 2G' = g(x)$$

$$\Rightarrow \begin{cases} F(x) + G(x) = f(x) & (1) \\ -2F'(x) + 2G'(x) = g(x) & (2) \end{cases}$$

Đạo hàm (1) $\Rightarrow F'(x) + G'(x) = f'(x)$ Thay vào (2), ta có

$$-2[f'(x) - G'(x)] + 2G'(x) = g(x)$$

$$\Rightarrow 4G'(x) = 2f'(x) + g(x)$$

$$\Rightarrow G'(x) = \frac{1}{2}f'(x) + \frac{1}{4}g(x)$$

$$\Rightarrow G(x) = \frac{1}{2}f(x) + \frac{1}{4} \int_0^x g(s) ds + C_1$$

$$\Rightarrow F(x) = \frac{1}{2}f(x) - \frac{1}{4} \int_0^x g(s) ds - C_1$$

$$\rightarrow u(x,t) = F(x-2t) + G(x+2t) \quad x+2t$$

$$= \frac{1}{2} [f(x-2t) + f(x+2t)] + \frac{1}{4} \int_{x-2t}^{x+2t} g(s) ds$$

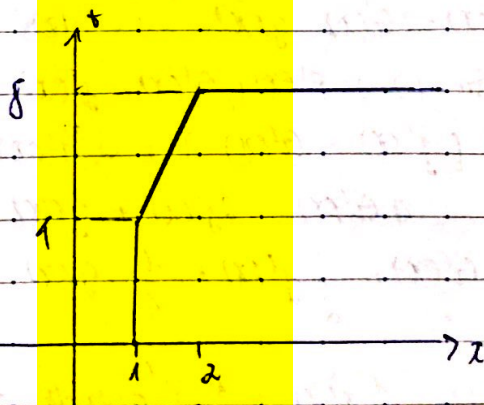
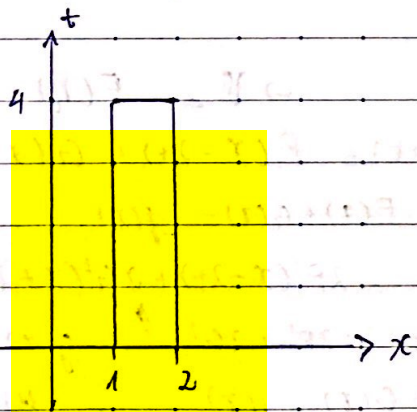
Điều kiện:

$$\begin{cases} u(x,0) = f(x) \\ u_t(x,0) = g(x) \quad (T/M) \\ u_{tt} = 4u_{xx} \end{cases}$$

Sóng tiến: $F(x) = \frac{f(x)}{2} - \frac{1}{4} \int_0^x g(s) ds$

Sóng lùi: $G(x) = \frac{f(x)}{2} + \frac{1}{4} \int_0^x g(s) ds$

$$g(x) = \begin{cases} 4 & \text{nếu } 1 < x < 2 \\ 0 & \text{nếu ngược lại} \end{cases}$$



$$\int_0^x g(u) du = \begin{cases} 0 & \text{nếu } x \leq 1 \\ 4x & \text{nếu } 1 < x < 2 \\ 8 & \text{nếu } x \geq 2 \end{cases}$$

$$f(x) = \begin{cases} 1-x^2 & \text{nếu } -1 < x < 1 \\ 0 & \text{nếu } x \leq -1 \text{ hoặc } x \geq 1 \end{cases}$$

$$\rightarrow F(x) = \frac{f(x)}{2} - \frac{1}{4} \int_0^x g(u) du = \begin{cases} 0 & \text{nếu } x \leq -1 \\ \frac{1-x^2}{2} & \text{nếu } -1 < x < 1 \\ -x & \text{nếu } 1 < x < 2 \\ -2 & \text{nếu } x \geq 2 \end{cases}$$

$$G(x) = \frac{f(x)}{2} + \frac{1}{4} \int_0^x g(u) du = \begin{cases} 0 & \text{nếu } x \leq -1 \\ \frac{1-x^2}{2} & \text{nếu } -1 < x < 1 \\ x & \text{nếu } 1 < x < 2 \\ 2 & \text{nếu } x \geq 2 \end{cases}$$



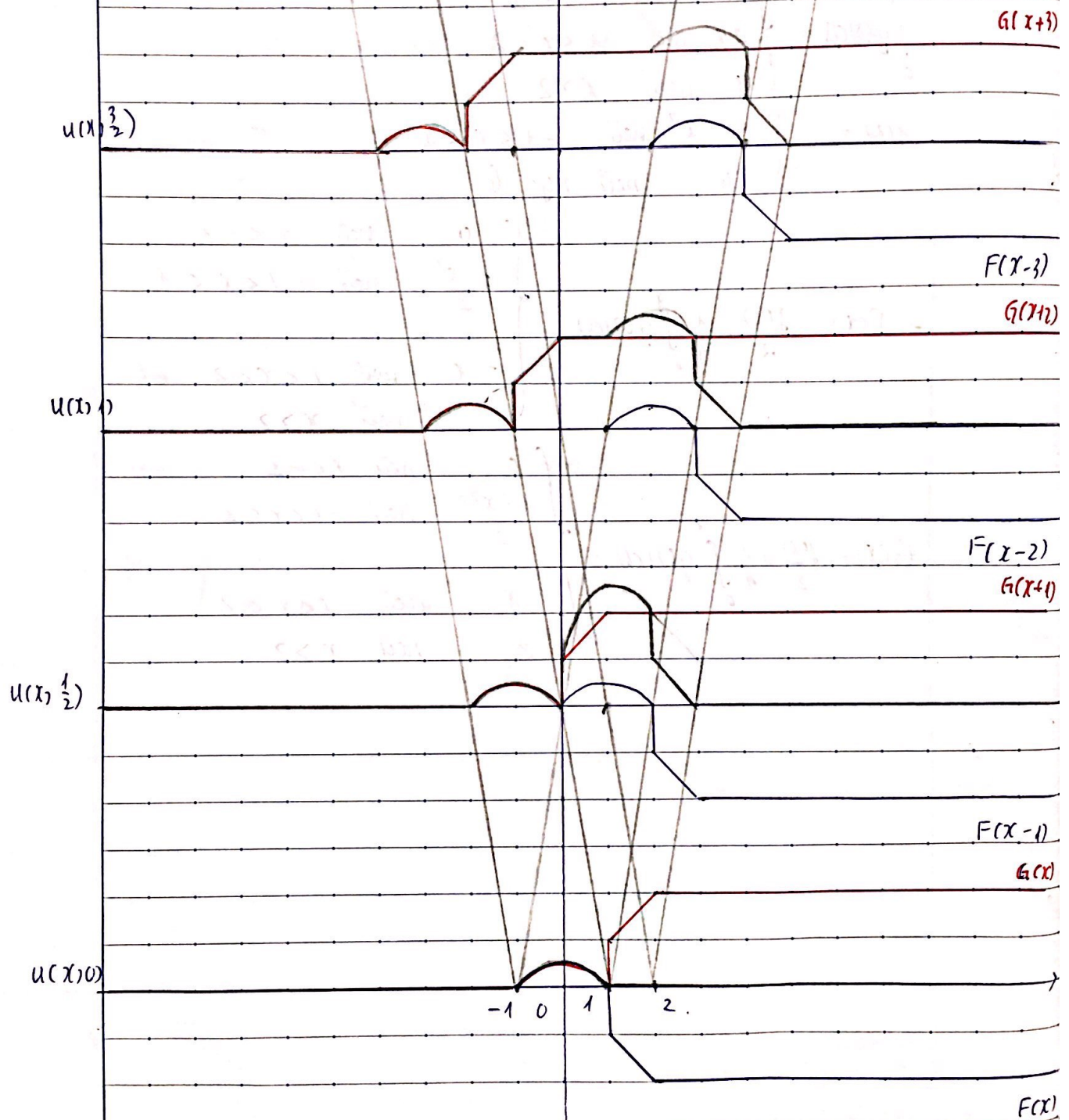
Thứ

No.

$$x+2t=-1 \quad x+2t=1 \quad x+2t=0$$

$$x-2t=-1 \quad x-2t=1$$

$$x-2t=2$$



Tập các đường lấy dữ của $u(x,t)$ gồm 6 tia.

$$\begin{cases} x+2t=+1 \\ x+2t=2 \\ x-2t=+1 \end{cases}$$

$$\begin{cases} x+2t=2 \\ x-2t=+1 \end{cases}$$

$$\begin{cases} x-2t=+1 \end{cases}$$

4.9

$$u_{tt} - u_{xx} = 1 \quad x \in \mathbb{R}, t > 0 \quad (*)$$

$$u(x, 0) = x^2 \quad x \in \mathbb{R}$$

$$u_t(x, 0) = 1 \quad x \in \mathbb{R}$$

Phương trình đặc trưng

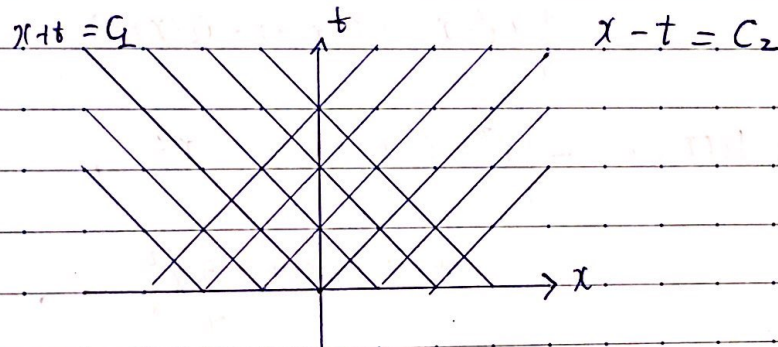
$$(dx)^2 - (dt)^2 = 0$$

$$\Rightarrow dx \pm dt = 0$$

 $\Rightarrow (*)$ là Hyperbolic

$$dx \pm dt = 0$$

$$\Rightarrow \begin{cases} x+t = C_1 \\ x-t = C_2 \end{cases}$$



Đổi biến $\begin{cases} \xi = x+t \\ \eta = x-t \end{cases} \rightarrow \begin{cases} x = \\ t = \end{cases}$

		v_ξ	v_η	$v_{\xi\xi}$	$v_{\xi\eta}$	$v_{\eta\eta}$
0	u_x	1	1	0	0	0
0	u_t	1	-1	0	0	0
-1	u_{xx}	0	0	1	2	1
0	u_{xt}	0	0	1	0	-1
1	u_{tt}	0	0	1	-2	1
Tổng	1	0	0	0	-4	0

→ Dạng chung của (1) là

$$-4v_{\xi\eta} = 1$$

$$\rightarrow v_{\xi\eta} = -\frac{1}{4}$$

$$\rightarrow v_{\xi} = -\frac{1}{4}\eta + g(\xi)$$

$$\rightarrow v = -\frac{1}{4}\eta\xi + F(\eta) + G(\xi)$$

$$\rightarrow u(x,t) = -\frac{1}{4}(x-t)(x+t) + F(x-t) + G(x+t)$$

$$= -\frac{1}{4}(t^2 - x^2) + F(x-t) + G(x+t)$$

$$u(x,0) = -\frac{1}{4}x^2 + F(x) + G(x) = x^2$$

$$\Leftrightarrow F(x) + G(x) = \frac{5}{4}x^2$$

$$u_t(x,t) = \frac{t}{2} + G'(x+t) - F'(x-t)$$

$$u_t(x,0) = G'(x) - F'(x) = 1$$

$$\rightarrow \begin{cases} u(x) & F(x) + G(x) = \frac{5}{4}x^2 & (1) \\ & -F'(x) + G'(x) = 1 & (2) \end{cases}$$

Đạo hàm (1)

$$\rightarrow F'(x) + G'(x) = \frac{5}{2}x$$

Thay vào (2), ta được

$$-(\frac{5}{2}x - G'(x)) + G'(x) = 1$$

$$\rightarrow 2G'(x) = \frac{5}{2}x + 1$$

$$G'(x) = \frac{5}{4}x + \frac{1}{2}$$

$$\rightarrow G(x) = \frac{5}{8}x^2 + \frac{x}{2} + C_1$$

$$\rightarrow F(x) = \frac{5}{4}x^2 - G(x)$$

$$= \frac{5}{8}x^2 - \frac{x}{2} - C_1$$

$$\rightarrow u(x,t) = \frac{1}{4}(t^2 - x^2) + F(x-ct) + G(x+ct)$$

$$= \frac{1}{4}(t^2 - x^2) + \frac{5}{8}(x-ct)^2 - \frac{x-ct}{2} - C_1 + \frac{5}{8}(x+ct)^2 + \frac{x+ct}{2} + C_1$$

$$= \frac{1}{4}(t^2 - x^2) + \frac{5}{8}[(x-ct)^2 + (x+ct)^2] + t$$

$$= \frac{1}{4}(t^2 - x^2) + \frac{5}{8}(2x^2 + 2t^2) + t$$

$$= \frac{1}{4}t^2 - \frac{1}{4}x^2 + \frac{5}{4}x^2 + \frac{5}{4}t^2 + t$$

$$= \frac{3}{2}t^2 + x^2 + t$$

Thử lại: $u(x,0) = x^2$ (T/M)

$$u_t(x,t) = 3t + 1 \rightarrow u_t(x,0) = 1$$
 (T/M)

$$u_{xx} = 2$$

$$u_{tt} = 3$$

$$\rightarrow u_{tt} - u_{xx} = 1 \text{ (T/M)}$$